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Evaluation of UK Broadband Subsidies

Abstract:
The UK's Superfast Broadband program was a major infrastructure initiative designed to eliminate the digital divide. This dissertation evaluates its aggregate impact on network quality over a 15 year horizon. Exploiting the program's staggered rollout and correcting for endogenous treatment selection via genetic matching, we apply a heterogeneity-robust difference-in-differences estimator. Using independent physical speed-test data, we find the intervention successfully expanded superfast coverage to become near universal in the UK. However, we also find evidence of a longer term "copper trap" - a policy-induced reduction in the deployment of faster 'Full Fibre' technology relative to untreated regions.

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1 Introduction

In 2015, the Prime Minister, David Cameron, called access to fast internet, "fundamental to life in 21st-century Britain," promising to deliver broadband with the same universal zeal as, "our forebears effectively brought gas, electricity, and water to all" (Cameron 2015). The OECD posits that substantial positive externalities arise from connectivity: for households, it reduces isolation and expands access to essential services; for firms, it manifests as agglomeration effects through remote employment and teleconferencing (OECD 2010). These benefits are received not just by the consumer but everyone in the wider network and so justifies subsiding broadband - particularly in areas where there is a lack of private provision (the so-called 'digital divide').

In that spirit, the government initiated the Superfast Broadband program to be led by Broadband Digital UK (BDUK). The project reported a highly favourable Benefit-Cost Ratio of 4.70, significantly outperforming flagship projects such as the Elizabeth Line, which reported a Benefit-Cost Ratio of 1.90(Transport for London 2024), or even typical road investments. However, this dissertation proposes that this is an overestimate, as it neglected longer-term lock-in costs. By operating under a "technology-neutral" mandate, BDUK awarded subsidies based purely on a 30 Mbps speed target. This allowed the incumbent to fulfill contracts using cheaper, legacy copper networks by building Fiber to the Cabinet (FTTC, delivering 30-80Mbps).

While this provided immediate upgrades, it saturated regions with an intermediate technology. By satisfying the baseline demand for connectivity, the state reduced the marginal willingness-to-pay for future upgrades. Deprived of this premium, it became unprofitable for new entrants to justify the costs of building Full Fibre (FTTP, delivering 1000+ Mbps) networks, effectively foreclosing the market. Conversely, this reduction in marginal revenue did not occur in untreated regions, new entrants could profitably deploy FTTP in those areas as demand grew. We define this lock-in as a "copper trap" — a market dynamic that would cause a new, policy-induced digital divide.

We test this hypothesis using a longitudinal, balanced panel of 201 UK Local Authorities observed over 15 years and source our dependent variables from a novel dataset of internet speed-tests. Our identification strategy exploits the staggered temporal and spatial rollout of the BDUK phases. We then apply the heterogeneity-robust estimator proposed by Sun and Abraham (2021) to recover dynamic treatment effects. Our results indicate that whilst BDUK subsidies achieved their objective of increasing Superfast broadband coverage, they simultaneously incurred a dynamic penalty. Four to six years post-intervention, treated local authorities exhibit a pronounced deficit in Full Fibre coverage relative to matched untreated counterparts. Finally, we run a continuous interaction model to isolate the proposed mechanism of a copper trap. We find a significant penalty coefficient of up to 9.2 percentage points for the most affected regions.

2 Background and Literature Review

2.1 Broadband Technologies

The capital expenditure and performance capabilities of broadband are dictated by the penetration depth of optical fibre relative to the end-user. Figure 1 demonstrates the three predominant network topologies in the UK market:

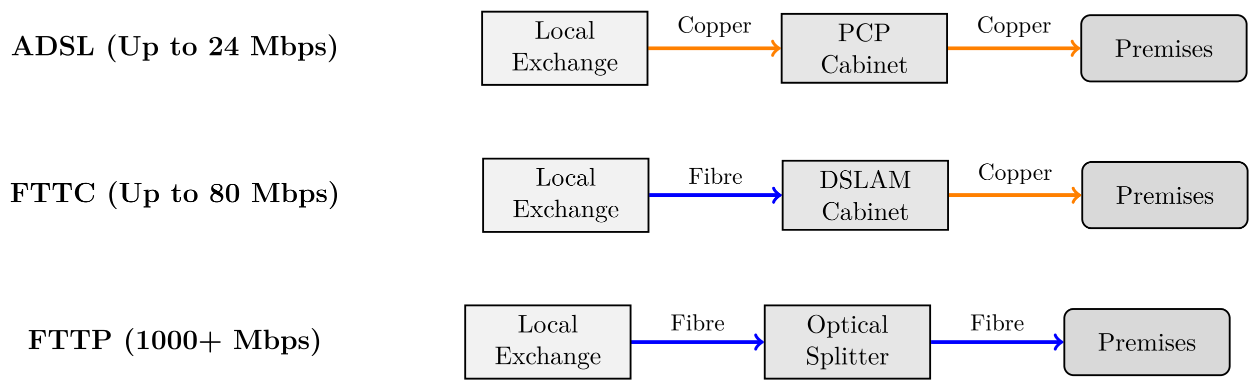


Figure 1: Comparative Broadband Network Topologies (ADSL, FTTC, and FTTP)

Each deployment method is successively more expensive.

Legacy Copper (ADSL) relies on copper lines from the local exchange to the home, resulting in severe distance-based attenuation and speeds capped at roughly 24 Mbps. These lines were built in the late '90s and remained a major technology through to 2010.

To achieve Superfast (>30 Mbps) speeds, the BDUK intervention predominantly subsidised Fibre-to-the-Cabinet (FTTC). This intermediate, low-CapEx technology upgrades the street cabinet and replaces the feeder segment with fibre. Crucially, by reusing the copper "last mile," the incumbent avoids civil trenching, though performance is capped at ~80 Mbps.

Conversely, Fibre-to-the-Premises (FTTP)—or "Full Fibre"—requires an upgraded optical splitter and end-to-end trenching to the home. This imposes a larger fixed-cost, costing approximately two to three times more than FTTC (Sky UK 2014). However, it creates a future-proof network capable of Gigabit (1,000+ Mbps) speeds with near-zero attenuation.

2.2 Market Structure and Regulation

Vertically integrated firms that control essential upstream resources have an incentive to deny competitors access to these inputs to protect downstream market power. This upstream monopoly allows the firm to 'foreclose competition' and distort consumer prices, as demonstrated by Hart and Tirole (1990). Consequently, policymakers must navigate the Schumpeterian trade-off between static efficiency (low prices) and dynamic efficiency (innovation and investment). Mandating "open access" to network resources promotes static efficiency but can compress the rent extraction required for private firms to incur the sunk fixed costs of Next Generation Access (NGA) upgrades to FTTC or FTTP.

Indeed, the UK broadband market was originally a state-led monopoly under BT. Following privatisation, the market evolved into a duopoly consisting of a legacy copper incumbent with universal reach (BT) and a cable-based competitor restricted largely to urban areas (Virgin Media). In 2006, following the separation of Openreach from BT, the UK adopted such a regulated open-access regime, meaning the Openreach network can be rented by any provider. While this drove down consumer prices through service-based competition, it reduced investment in NGA upgrades (Nandotto 2024).

This aligns with the results of the Klumpp and Su (2010) model, where an upstream monopolist chooses network quality investment alongside a Cournot oligopoly in the downstream market. They suggest that 'revenue-neutral' access pricing, similar to the frameworks set on Openreach, do encourage an incumbent to invest in quality. However, this is conditional on demand uncertainty being low. In rural regions of the UK, broadband take-up is uncertain, so sharing requirements dampen the incentive to upgrade due to asymmetric risk-sharing. For the incumbent, the investment in civil engineering constitutes a non-recoverable sunk cost; they bear 100% of the downside if take-up is zero, but under an open-access regime, they must share the gains with entrants if the investment is successful.

Klumpp and Su's model focuses solely on intra-platform competition, where service-based entrants operate on a single shared network. In practice, the UK also features inter-platform competition, primarily from the Virgin Media cable network and, increasingly, smaller fibre 'AltNets'. Research by Distaso et al. (2006) and Höffler (2007) provides evidence across Europe that inter-platform competition is a stronger catalyst for infrastructural upgrades than intra-platform service competition. Rather, intra-platform competition may incentivise incumbents to engage in "asset sweating"—maximising returns on legacy networks rather than upgrading them. (Bouckaert et al. 2010)

2.3 Subsidy Design and the BDUK Program

The consensus of the telecommunications literature is that broadband subsidies are welfare-enhancing when targeted at "white areas" — regions with zero NGA provision, as the positive externalities of broadband adoption are entirely absent.(Qiang and Rossotto 2009). Likewise, Jullien et al. (2010) support subsidies in these areas but emphasise that intervention in 'black' or 'gray' areas, that have at least one operator, is likely welfare-reducing. In such regions, marginal benefits shrink and costs can be inflated due to asymmetric information, as the state lacks precise data on a firm's true capital costs, creating a principal-agent problem during negotiations.

By 2010, the UK had many such 'white areas' as its NGA rollout lagged behind other OECD nations; only 0.2% of UK households possessed NGA connections, compared to 34% in Japan and 12% in Sweden (Ofcom 2010). Domestic reviews highlighted a persistent 'digital divide' in rural areas, where high engineering costs and low premise density rendered cabinet upgrades unprofitable (Department for Culture, Media and Sport (DCMS) 2010). To close this gap, the government initiated the Superfast Broadband program, led by BDUK, in three phases:

- Phase 1 (2010–2015): £530m to reach 90% Superfast coverage.
- Phase 2 (2015–2017): £250m to extend Superfast coverage to 95%.
- Phase 3 (2016–2020): A shift toward a "Gigabit-capable" baseline for the final 5%.

BDUK utilised a "gap-funding" model where providers were awarded a subsidy to bridge the deficit between project costs and normal returns. This introduced a significant cost asymmetry: while the incumbent could cheaply upgrade its legacy copper to FTTC, entrants were restricted to leasing its lines, meaning a rival FTTC bid required constructing redundant infrastructure—a proposal easily undercut by BT (Bourreau et al. 2012; Sky UK 2014). Consequently, alternative providers were forced to put in proposals using capital-intensive Full Fibre (FTTP). These were systematically outbid by the incumbent's cheaper FTTC proposals under the BDUK's technology-neutral review process, regardless of the difference in promised speeds. This resulted in BT winning every contract in Phases 1 and 2 (Ipsos MORI 2018), validating the critique by Briglauer and Holzleitner (2014) that ex-ante speed targets inherently favours incumbents.

Finally, a well-designed subsidy should account for the temporal dynamics of firm entry. Applying real options theory, Dixit and Pindyck (1994) demonstrate that firms treat sunk investments in uncertain markets as call options, rationally delaying deployment until demand uncertainty resolves. Consequently, static white areas may simply reflect strategic deferral. State subsidies that fail to account for this may inadvertently crowd out organic private investment that would have eventually materialised as market viability improved over time.

2.4 Government Evaluations

Existing evaluations of the BDUK program have predominantly focused on short-term macroeconomic impacts and basic Superfast take-up. The first official evaluation established that the program successfully expanded basic coverage without breaching competition laws (Oxera 2013). Subsequent appraisals covering Phases 1 and 2 mapped localised impacts, such as on employment growth and firm turnover (Ipsos MORI 2018), while the final reports quantified the program's cumulative value (Ipsos 2023).

These institutional reports share two critical limitations: they rely on operator's coverage claims rather than real sampled broadband speeds, and they evaluate static outcomes while ignoring longer-term impacts on infrastructural competition.

This dissertation addresses both issues by not just improving the data being used, but also by providing a theoretical contribution. While the government evaluations document 'what' happened during the program, they do not explore the underlying mechanics. By applying a simple but formal model of the broadband market, this dissertation proposes a mechanism that can explain 'why' the initial policy design in Phases 1 and 2 drove the market—as observed empirically—into a "copper trap."

3 Conceptual Framework

To formalise the empirical hypothesis, we model the broadband market as a two-period sequential game under asymmetric costs. We use the same theoretical framework established by Bourreau et al. (2012), but further introduce positive externalities.

3.1 Model Setup

We define the rules of the game as follows, and visualise it below in Figure 2.

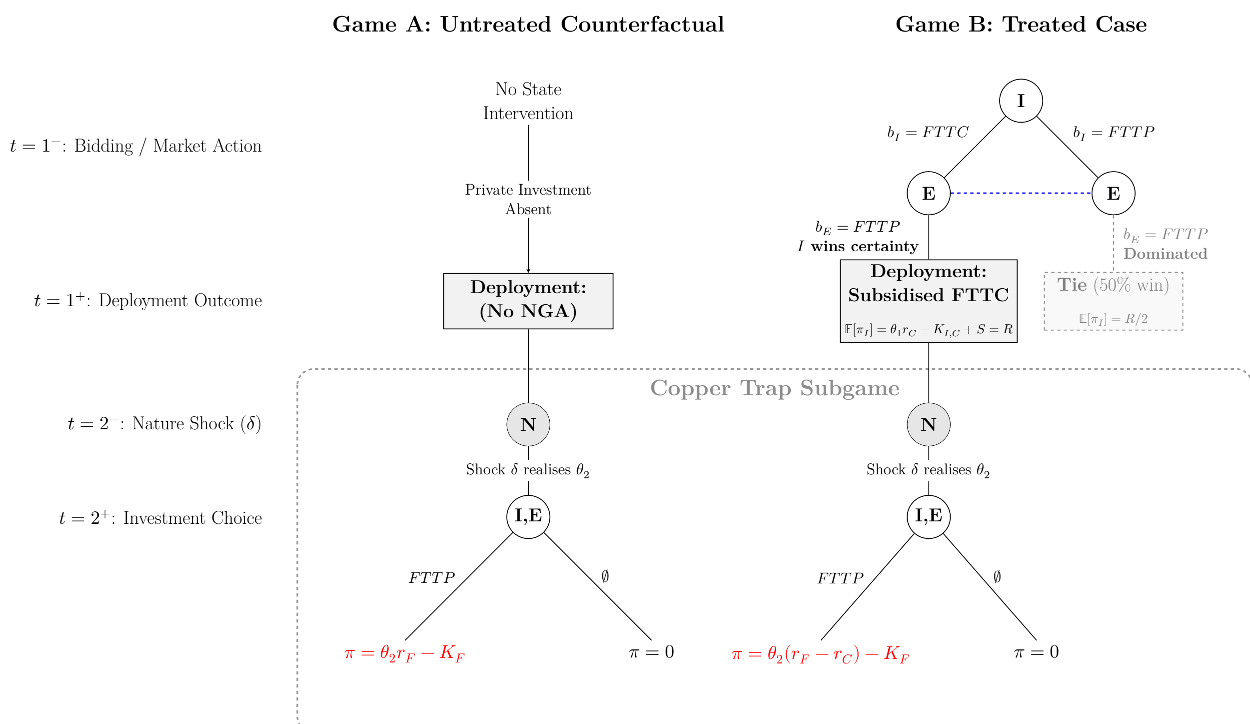


Figure 2: Parallel Post-Shock Investment Subgames. The red highlights contrast absolute monopoly returns in untreated areas (2A) with the compressed returns to deploying FTTP in treated areas (2B)

Environment: The market comprises a continuum of local telephone exchanges, indexed by j. At time t, each exchange is characterised by a commercial viability score $\theta_{j,t} \in [0, \infty)$, which captures unit-specific supply and demand shifters (e.g., premise density, household income).

Players: The state is an agent (S) seeking to minimise subsidy outlays subject to a baseline quality constraint ($q \geq q_C$); the firms are profit-maximising agents with Incumbent (I) possessing a copper network; and an Entrant (E) with no existing assets.

Technologies: Two NGA technologies exist: FTTC (quality q_C) and FTTP (quality q_F), where $q_F > q_C$. FTTC inherently requires a relatively minor capital cost ($K_{I,C}$). FTTP requires end-to-end trenching, imposing a large fixed cost such that $K_F \gg K_{I,C}$.

Strategies: Due to its monopoly over legacy assets, the Incumbent has the exclusive option to deploy FTTC. Therefore, Incumbent's strategy space is either deploying FTTC, deploying FTTP, or do nothing. The Entrant lacks legacy assets, its strategy space is strictly limited to deploying FTTP or do nothing.

Payoffs: A firm's payoff (π) is the profit that it earns in a market, this profit is dependent on the existing infrastructure.

- Unserved Market: If no NGA infrastructure exists, a firm captures the absolute monopoly revenue for technology k, yielding $\pi = \theta r_k - K_k$ where $k \in \{C, F\}$. We assume $r_F > r_C$ as consumers exhibit a higher marginal willingness-to-pay for faster speeds.
- Saturated Market: If an FTTC network is already deployed, then one can only subsequently deploy FTTP. Whether it is the Incumbent cannibalising its own network or a new Entrant building over it, that firm's profit is restricted to the marginal willingness-to-pay so: $\pi = \theta(r_F - r_C) - K_F$.

Externalities: Broadband generates positive societal externalities (e_j), where $e_F \gg e_C > 0$, which private agents ignore, but they justify the State's intervention.

Timing: Finally, in terms of timing the game unfolds as follows.

- $t = 1^-$ (Policy Intervention): The State identifies a "White Area" where θ_1 is too low for private investment. Under a technology-neutral mandate, S commits to subsidising the lowest-cost bid that clears the q_C threshold.
- $t = 1^+$ (Procurement & Deployment): Firms submit bids. Because the State prioritises immediate cost-minimisation, the Incumbent's profit-maximising FTTC bid is selected over any FTTP bidders due to the cost advantage ($K_{I,C}$). FTTC is deployed.
- $t = 2$ (Future Entry): The demand shock is realised ($\theta_2 = \theta_1 + \delta$). The Entrant observes the deployed $t = 1^+$ infrastructure and decides whether to organically enter the market with FTTP.

3.2 Social Planner's Solution (First-Best)

To establish a benchmark against which market outcomes may be evaluated, we first find the first-best optimum. A socially benevolent planner maximises total welfare $W_j(\theta)$; the sum of private revenue and positive externalities, net of capital costs.

$$W_j(\theta) = \theta(r_j + e_j) - K_j$$

The Socially Optimal Threshold (θ_F^{SP}) for FTTP deployment occurs where the welfare of FTTP exceeds FTTC ($W_F \geq W_C$). Because private firms disregard societal externalities (e_F), the free-market entry threshold for FTTP (θ_F^{priv}) will strictly exceed the social planner's optimal threshold:

$$\theta_F^{SP} = \frac{K_F - K_{I,C}}{(r_F + e_F) - (r_C + e_C)} < \theta_F^{priv} = \frac{K_F - K_{I,C}}{r_F - r_C}$$

Thus, any authority lying between $\theta \in \theta_F^{SP}, \theta_F^{priv}$, is a white area that should qualify for *full fibre* broadband subsidies.

3.3 Subgame Perfect Nash Equilibrium (Second-Best)

We add a secular demand shock ($\theta_2 = \theta_1 + \delta$) between $t = 1$ and $t = 2$ to capture the organic growth of usage improving commercial viability over time.

Solving the sequential game via backward induction reveals how state intervention at $t = 1$ alters the investment subgame at $t = 2$. If the State intervenes at $t = 1$ to subsidise a designated "White Area", the singular focus of cost-minimisation mean it will strictly select the Incumbent's FTTC bid, given the profound cost asymmetry. At $t = 2$, should a prospective Entrant desire to construct an FTTP network, it must confront a market already saturated with state-sponsored FTTC. To win customers, the Entrant can no longer extract the monopoly revenue (r_F); rather, it is constrained to capturing merely the marginal willingness-to-pay for the incremental speed upgrade ($r_F - r_C$). The full solution lies in Appendix B.

We map the resulting SPNE across the spectrium of local authorities in t=2, (θ_2) for three scenarios: (i) where there is a 1st best allocation, (ii) a 2nd best allocation with no intervention and (iii) a 2nd best allocation but with state intervention in t=1 to provide universal superfast coverage in Figure 3.

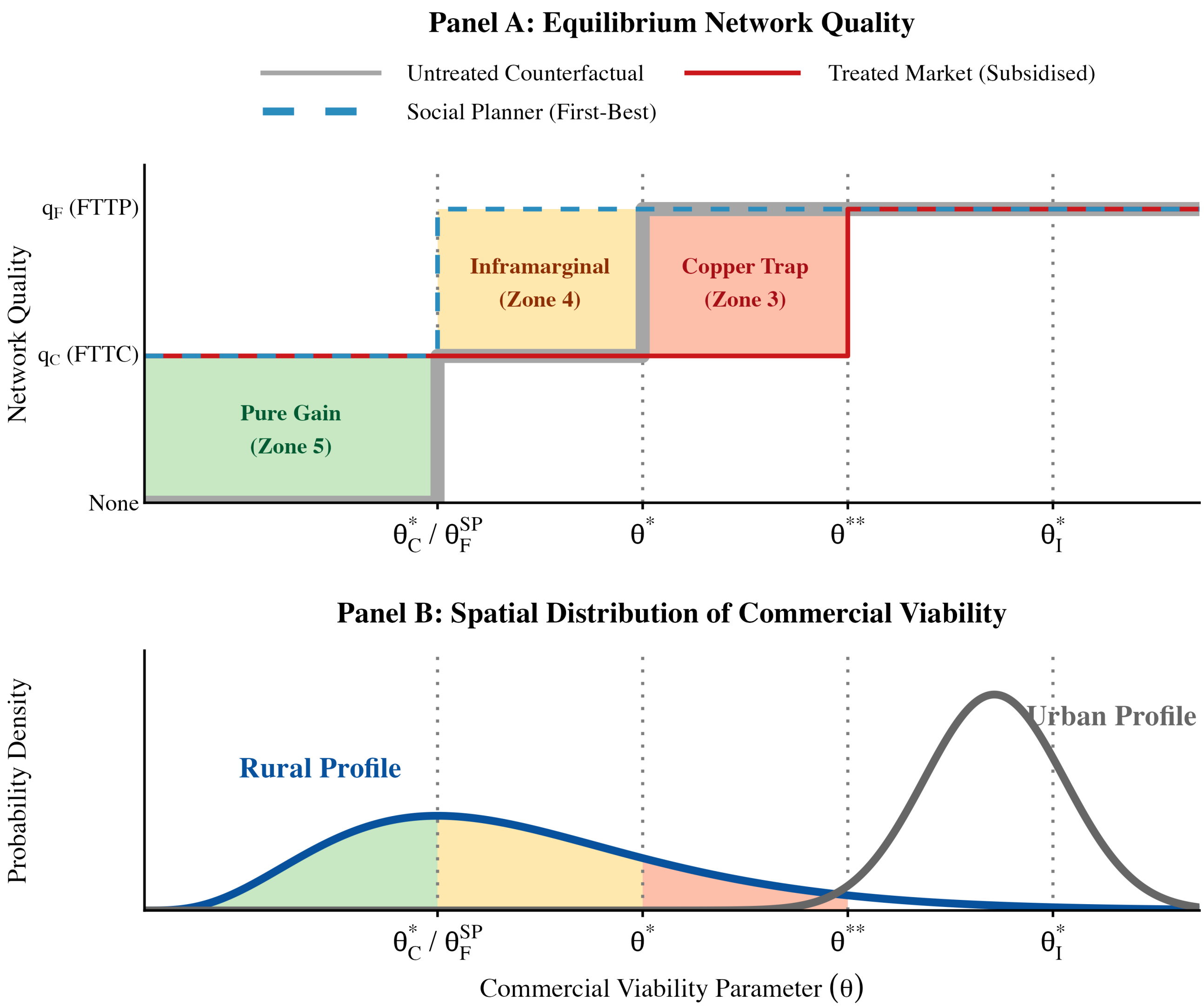


Figure 3

Interpreting the Zonal Equilibrium

Zone 5 (Pure Gain): At the lowest bound of viability ($\theta_2 < \theta_C^*$), the market is characterised by severe failure. Even following the demand shock, the region remains too sparsely populated to support private full fibre. In the untreated counterfactual, this market remains entirely unserved. Consequently, the state's initial intervention to subsidise FTTC yields a pure static welfare gain. It successfully provides fast internet connectivity to a market that the private sector would never have reached.

Zone 4 (Inframarginal): As viability increases into the subsequent interval ($\theta_2 \in [\theta_C^*, \theta^*)$), the demand shock proves sufficiently robust that the incumbent would have organically upgraded to FTTC in the second period. In this scenario, the subsidy is inframarginal and not the best use of public funds, though there is still a welfare gain as the infrastructure is supplied earlier at $t = 1$, allowing society to capture positive externalities earlier than if the private market would have delivered them.

Zone 3 (The Copper Trap): The friction emerges within the interval $[\theta^*, \theta^{**})$. Here, intrinsic viability has matured sufficiently to support an organic FTTP network in an untreated counterfactual. However, the prior introduction of state-sponsored FTTC artificially elevates the entry barrier to θ^{**}. This revenue compression means the initial state intervention forecloses the deployment of full fibre in the future. The region is deprived of the larger positive externalities (e_F) associated with full fibre because it is still stuck in FTTC, resulting in a welfare loss.

Zone 2 (Not Subsidised): Beyond this threshold, in the interval $[\theta^{**}, \theta_I^*)$, are the markets that are highly lucrative. New entrants can profitably overbuild the state-sponsored FTTC network with its own FTTP network.

Zone 1 (Not Subsidised): Finally, in hyper-dense areas ($\theta_2 \geq \theta_I^*$), market density is so extreme that the incumbent itself abandons the extraction of rents from its legacy assets, choosing to go straight to FTTP, even in $t = 1$.

The focus of this paper is on the local authorities bounded by the Overbuild Threshold (θ^{**}). Our aim is to test this 'copper trap' hypothesis empirically by evaluating whether the theorised difference in FTTP rollout between treated and untreated regions within this zone are actually present in the data.

4 Data and Empirical Framework

4.1 Dataset Construction

To evaluate the long-term impact of the BDUK intervention, we construct a longitudinal, balanced panel of 201 UK Unitary and County Local Authorities observed annually over a 16-year period from 2010 to 2025 (N = 3,216).

A primary issue with past evaluations completed by BDUK or government-commissioned economic consultants is their reliance on statutory regulatory data from Ofcom. Ofcom's broadband coverage reports rely on maximum speeds that are advertised, modelled and submitted by the incumbent providers themselves. This is known to be optimistic (Department for Culture, Media and Sport (DCMS) 2010). To address this, our dependent variables—fractional Superfast Coverage ($y_{SF,it}$) and Full Fibre Coverage ($y_{FF,it}$)—are sourced from ThinkBroadband thinkbroadband.com (2026). This independent commercial vendor tracks network footprints from thousands of visitors to its speed-test website, and corroborates this with direct infrastructural mapping. Their rich dataset ensures our outcomes reflect genuine, on-the-ground network capabilities (see Appendix A for a full comparison between Ofcom and ThinkBroadband datasets).

We merge this data with annual ONS statistics (Office for National Statistics 2025b, 2025a) to capture the explanatory variables of commercial viability: Gross Disposable Household Income (GDHI), population density, and the proportion of residents aged over 65. From a fundamental standpoint, these are core determinants of a region's rollout costs and consumer willingness-to-pay. From a practical standpoint, they are high-quality, consistently observable data at the local authority level. They are also not susceptible to reverse causality - so they serve as ideal covariates for matching (discussed in Empirical Framework).

During the construction of this panel, extensive data cleaning was required and resolved three main challenges. First, to resolve naming inconsistencies across the 16-year period, authority names were standardised to account for Welsh and Gaelic variations. Second, to prevent artificial jumps in broadband coverage caused by municipal boundary changes (for instance, Dorset County Council was merged with four other areas in 2019), all data were strictly mapped to static 2010 Local Authority definitions. Finally, because BDUK partnered exclusively with 'unitary authorities', the dataset was stripped of all lower-tier district councils. This ensured a consistent level of observation, resulting in the final balanced panel of 201 unitary authorities. After this, the data was checked for completeness, and consistency (e.g., identifying anomalous decreases in speed or coverage metrics for next-generation access or fibre). The data was further validated to ensure that absence of data related to gigabit speeds was due to their introduction mid-period (i.e., 2017, 2018, or later) rather than data quality issues.

4.2 Treatment Timing

Funding allocation dates were sourced from public FOI requests and official BDUK audit reports. `Implementation_Year` is defined as the year the Local Authority hit its first milestone (M1) of bringing subsidised builds online. The treatment variables, T_{early} (Phases 1 and 2) and T_{late} (Phase 3), activate in the implementation year; naturally this implementation year is staggered, motivating use of a staggered DiD. The distinguishing of T_{early} and T_{late} was considered necessary given the style shift in the awards for Phase 3.

This is summarised as the map in Figure 4:

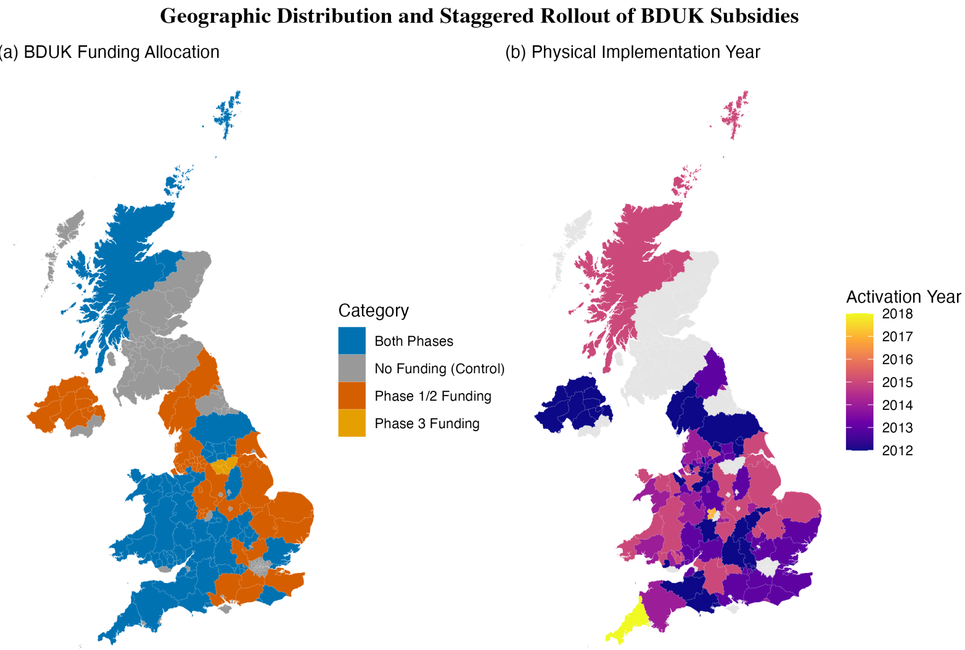


Figure 4: Geographic Distribution and Staggered Rollout of BDUK Subsidies

The spatial and temporal heterogeneity of the program is illustrated in Figure 4. Panel (a) maps the funding classifcations; about half received *both* Phase 1/2 and Phase 3 funding, while the other half only received Phase 1/2 funding. Panel (b) maps the actual year of implementation. Due to the commercial sensitivity of exact exchange-level data, we proxy the onset of treatment using the 'M1' milestone, which records when 20% of a Local Authority's planned cabinets become active, from BDUK audit reports, While this introduces temporal fuzziness, subsequent placebo tests demonstrate that this error safely attenuates our estimates toward zero.

Key to note is the "checkerboard" pattern across the UK in panel (b). This visually confirms the highly staggered nature of the rollout, a dynamic that invalidates standard panel estimators and motivates our use of a Sun and Abraham (2021) event study design.

4.3 Summary Statistics

Aggregate statistics reveal that both Superfast and Full Fibre broadband have transitioned from zero coverage to broad ubiquity over the last 16 years. Indeed, looking at the raw distribution of Full Fibre gains between 2010 and 2025, Figure 5, both treated and control areas registered immense improvements over +70 percentage points.

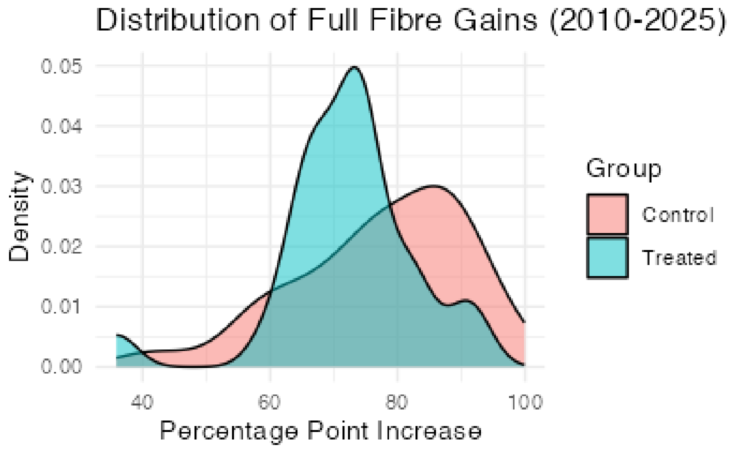


Figure 5: Distribution of Full Fibre Coverage Gains (Treated vs Control)

However, a distinct right-tail skew exists for untreated groups, meaning untreated regions still ended up building out more full fibre. A naive comparison of these group averages would be confounded by selection bias: treated regions were targeted *precisely* because they suffered from lower commercial viability. To isolate the true impact of the state intervention, we must strip away the selection effect.

Table 1 presents the pre-treatment baseline characteristics for our sample in 2010. Observing the 'Difference' column and associated p-values reveals imbalances between the treated and control groups prior to matching. For instance, the control group possessed a significantly larger legacy Virgin Media footprint (58% vs 26%, $p < 0.001$) and nearly double the population density. The significant p-values confirm treated regions were disadvantaged at baseline.

If left uncorrected, these would bias any standard panel estimator. The following empirical framework addresses this issue.

Table 1: Summary Statistics (Pre-Treatment Baseline, 2010)

Variable	**Control Group** ($N = 896$)	**Treated Group** ($N = 976$)	**Difference** ($p - value$)
Superfast Coverage (%)	0.14 (0.06)	0.10 (0.04)	0.001
Full Fibre Coverage (%)	0.01 (0.01)	0.01 (0.01)	0.842
GDHI per head (£)	19,100 (3,500)	17,800 (2,900)	0.021
Pop. Density (km^2)	580 (150)	320 (90)	<0.001
Residents 65+ (%)	17.5 (2.1)	18.9 (2.6)	0.045
Virgin Media (%)	0.58 (0.30)	0.26 (0.22)	<0.001

Note: Baseline values calculated for 2010. Standard deviations in parentheses. P-values derived from standard t-tests.

4.4 Empirical Framework

Estimating the causal effect of broadband deployment is complicated by endogenous selection into the program, the staggered timing of infrastructure rollout, and dynamic panel effects. Because BDUK funding explicitly targeted commercially unviable "White Areas," a naive comparison would conflate the treatment effect with underlying geographic and economic disparities.

In a panel data setting with fixed effects, the Conditional Independence Assumption (CIA) requires that there are no time-varying, unit-specific confounders. If this holds, treated and untreated units, conditional on the covariates, would have followed parallel trends in the absence of the intervention. To make this assumption credible and establish a valid counterfactual, we correct for baseline selection bias by deploying 1:1 genetic matching with a 0.1 caliper. We match Local Authorities on the invariant determinants of commercial viability: population density, age, income, exchange density, and legacy Virgin Media footprint (y_{vm}). We restrict our matching matrix to characteristics that are strictly exogenous to the outcome variable to prevent post-treatment bias. For instance, the Virgin Media network serves as an invariant supply-side constraint; built upon historical coaxial cable laid in the 1990s, its footprint was fixed two decades prior to the BDUK intervention. It therefore acts as a strictly exogenous proxy for the baseline 'ease of installation', rather than a variable responding to contemporaneous fibre deployment. Likewise, macroeconomic and demographic factors such as age, income, and population density are stable and not susceptible to reverse causality from the outcome variable.

Our selection of covariates follows the precedent set by official government appraisals, which relied on these same metrics to construct counterfactuals via Propensity Score Matching (PSM) (Ipsos MORI 2018). However, PSM rests on a fragile assumption: it forces covariates through a logistic regression to generate a single scalar score. If this underlying model fails to precisely capture complex non-linearities or interactions, the resulting match can remain unbalanced.

That is why we, rather than imposing a rigid functional form, we deploy Genetic Matching (Diamond and Sekhon 2013). This approach uses an iterative search to directly and nonparametrically maximise covariate balance between our treatment and control groups. Admittedly, the immense computational burden of this algorithm rendered it unfeasible for government reports evaluating millions of individual postcodes. However, our panel is aggregated at Local Authorities, which radically reduces the dimensionality of the data, affording us the luxury to do genetic matching.

By pruning hyper-urban outliers (e.g., London boroughs) that would severely violate the parallel trends assumption, our protocol isolates genuine local authority twins. For example, the algorithm successfully paired Cornwall—a sparse, highly rural unitary authority that served as a flagship Phase 1 BDUK recipient—with Herefordshire, an untreated authority sharing near-identical baseline characteristics: zero historic Virgin Media footprint, a matching proportion of over-65s, and similarly low Gross Disposable Household Income. As demonstrated in Figure 6, this procedure reduces the estimation sample from 1,872 to 1,378 observations and drives the Standardised Mean Differences (SMD) across our matched groups toward zero.

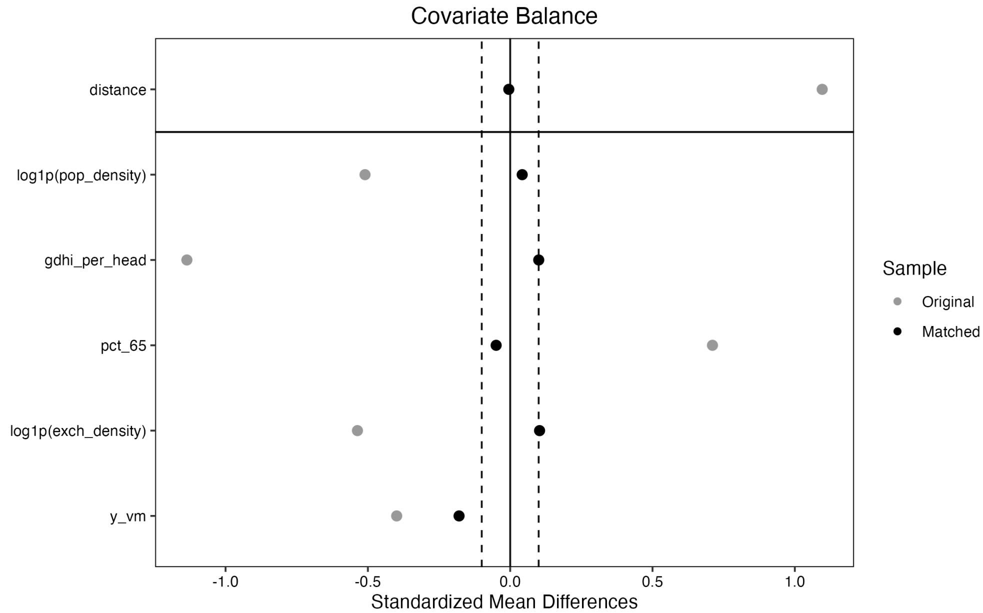


Figure 6: Covariate Balance (Genetic Matching)

Furthermore, we can rule out spatial spillovers—which would violate the Stable Unit Treatment Value Assumption (SUTVA) and contaminate the control group—because physical broadband infrastructure is constrained by hard telephone exchange boundaries rather than permeable administrative borders. BDUK funding was strictly ring-fenced at the cabinet level; because physical cabinets are wired to specific postcodes, the technological treatment cannot "bleed" into untreated control regions.

Having established a balanced sample free of spillover effects, we address the bias inherent in the staggered rollout of BDUK funding. Recent econometric literature (Goodman-Bacon 2021) demonstrates that in the presence of treatment effect heterogeneity, standard Two-Way Fixed Effects (TWFE) estimators apply negative weights to already-treated units, severely biasing the Average Treatment Effect on the Treated (ATT). To recover unbiased estimates, we deploy the Sun and Abraham (2021) interaction-weighted estimator on our matched sample. This method isolates the Cohort Average Treatment Effect on the Treated ($CATT = \delta_{e,l}$):

$$y_{it} = \alpha_i + \tau_t + \sum_{e} \sum_{l \neq -1} \delta_{e,l}(1\{E_i = e\} \times 1\{t - E_i = l\}) + X_{it}\gamma + \epsilon_{it}$$

where:

- y_{it} is broadband coverage (either SF or FF) for Local Authority i in year t.
- α_i and τ_t represent Local Authority and year fixed effects.
- E_i denotes the treatment cohort (the year Local Authority i first receives funding), making e the specific cohort group and l the relative time since treatment ($t - E_i$).

- $\delta_{e,l}$ is the Cohort Average Treatment Effect on the Treated ($CATT$) for cohort e at relative time l.
- X_{it} is a vector of covariates (income, age etc.)
- ϵ_{it} is the idiosyncratic error term.

To recover the overall Average Treatment Effect on the Treated (ATT_l), for a given relative year, the cohort estimates are aggregated using their sample shares ($w_{e,l}$) as weights:

$$ATT_l = \sum_e w_{e,l}\hat{\delta}_{e,l}$$

Finally, following Abadie et al. (2022), standard errors in all specifications are clustered at the Local Authority tier to account for serial correlation within units because standard OLS errors falsely assume all observations are independent, which ignores within-unit auto-correlation common in panel data.

While the Sun and Abraham (2021) event-study framework successfully quantifies the aggregate policy treatment effect, it remains agnostic to the underlying economic mechanism. Earlier, we hypothesised a "Copper Trap" channel where an existing stock of FTTC reduces new Full Fibre deployments. To test if this proposed channel holds, we estimate a continuous interaction Difference-in-Differences model:

$$y_{FF,it} = \alpha_i + \lambda_t + \beta_1 Post_{it} + \beta_2 y_{SF,i,t-1} + \beta_3(Post_{it} \times y_{SF,i,t-1}) + X'_{it}\gamma + \epsilon_{it}$$

where:

- $y_{FF,it}$ is the dependent variable representing Full Fibre coverage for Local Authority i in year t.
- $Post_{it}$ is a binary indicator equalling 1 if Local Authority i is in a post-treatment period (funding active), and 0 otherwise.
- $y_{SF,i,t-1}$ is the continuous interaction variable capturing the lagged stock of FTTC coverage from the previous year.
- β_1 isolates the baseline treatment effect of BDUK funding in a market with zero existing FTTC infrastructure.
- β_2 captures the independent effect of lagged FTTC coverage on Full Fibre rollout.
- β_3 is the parameter of interest: the marginal interaction penalty reflecting the disincentive to overbuild (the Copper Trap).
- α_i and λ_t denote Local Authority and year fixed effects, respectively.

Because estimating continuous treatment interactions with the Sun and Abraham (2021) framework is intractable, we revert to Two-Way Fixed Effects (TWFE). This is biased; we later argue why this should be small in Figure 10.

5 Results

5.1 Divergence in Network Speeds

Table 2: TWFE and S&A Estimates for Superfast and Full Fibre Coverage

	SF: TWFE (Mat)	SF: S&A (Mat)	FF: TWFE (Mat)	FF: S&A (Mat)
Post (Phase 1/2)	0.09^{**}		-0.04^{**}	
	(0.03)		(0.02)	
Post (Phase 3)	0.03^{+}		0.03	
	(0.02)		(0.02)	
GDHI per head (£)	-0.00	0.00	0.00	0.00
	(0.01)	(0.01)	(0.01)	(0.01)
Residents Aged 65+ (%)	11.46^{***}	11.71^{***}	-5.82^{***}	-4.16^{***}
	(1.29)	(1.32)	(1.38)	(0.87)
Exchange Density	-1.35^{**}	-1.57^{*}	-1.52	-1.34
	(0.50)	(0.63)	(1.39)	(0.91)
Post (S&A Average CATT)		0.02		-0.00
		(0.03)		(0.01)
Num.Obs.	1166	1484	1166	1484
R-squared	0.796	0.835	0.786	0.789
Local Authority Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Cluster SE (Local Authority)	Yes	Yes	Yes	Yes

$^{+}p < 0.1, ^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001$

S&A event-time coefficients omitted. GDHI, Density and %65+ are covariates in the vector X_t

Table 2 reports the aggregate impact of the BDUK intervention on 30Mbps+ Superfast (SF) and 1000Mbps+ Full Fibre (FF) coverage. Columns (1) and (3) use a Two-Way Fixed Effects (TWFE) estimator, where the 'Post' indicators measure the average difference in coverage between treated and control areas for the early and late phases of the program. Columns (2) and (4) provide the Aggregate Average Treatment Effect on the Treated (ATT) using the Sun and Abraham (2021) estimator, which represents the sample-weighted average of the cohort-specific effects ($\delta_{e,l}$) across all post-treatment years on SF and FF, respectively.

Table 2 provide the first evidence of a divergence in outcomes. In the TWFE specification, the early phase of the program is associated with a significant 9 percentage point increase in Superfast coverage ($p < 0.01$) but a simultaneous 4 percentage point decrease in Full Fibre deployment ($p < 0.01$). But note, the S&A specification's ATT for both outcomes is insignificant. This suggests the policy impact is not a permanent shift, rather the treatment effect varies over time (l) before reverting to zero, we use event study plots to visualise this.

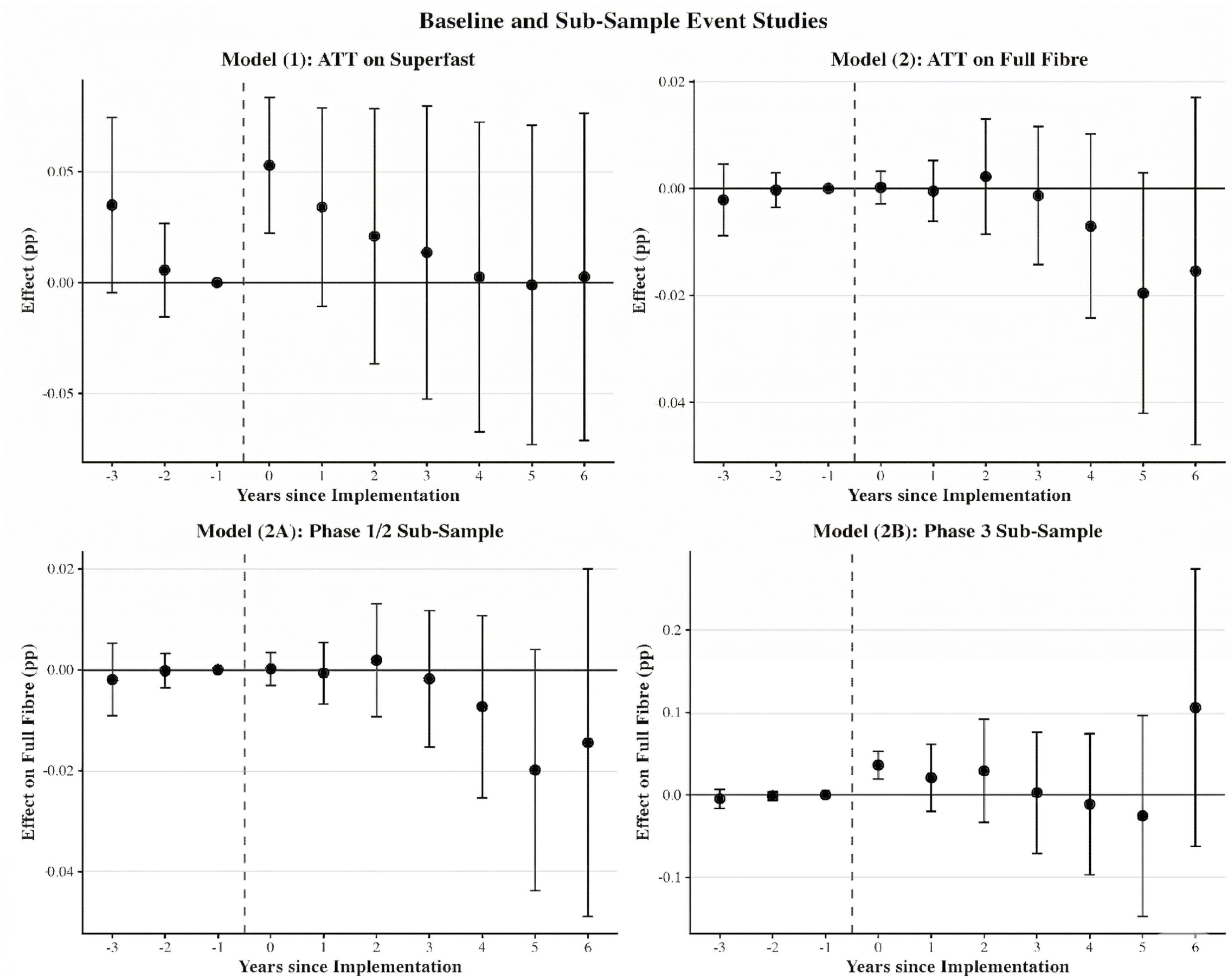


Figure 7: Baseline and Sub-Sample Event Studies

In Figure 7, each point represents a treatment effect coefficient from the interaction-weighted model. These coefficients are expressed in percentage points relative to the year prior to treatment ($T = -1$). A result is statistically significant if the 95% confidence interval bars do not cross the zero horizontal axis.

The Figure 7 event study reveals that the BDUK programme was highly successful in its primary objective. Model (1) demonstrates that the year where the new cabinets go online ($T = 0$), Superfast coverage experiences a highly significant, sustained jump of approximately 5 percentage points relative to the control group. The pre-treatment coefficients ($T \leq -1$) are insignificant at the 5% level, implying parallel pre-trends between treated areas and matched controls prior to intervention.

Evaluating this same intervention against the deployment of Full Fibre in Model (2A) reveals the impact of the Phase 1/2 Superfast intervention on Full Fibre (y_{FF}) coverage. While pre-trends remain parallel, the coefficients at $T = 4$ to $T = 6$ drop into the negative and become significant. Four to six years after the state subsidised a Superfast cabinet, that exact same Local Authority possesses a visible deficit in Full Fibre compared to a matched twin county that never received the subsidy.

A separate sub-sample analysis of only Phase 3 cohorts (Model 2B) indicates the foreclosure effect was not uniform. BDUK shifted away from technology neutrality in Phase 3. We find the Phase 3-treated cohorts avoided the Full Fibre deficit, experiencing instead a significant increase in FF coverage beginning at $T = 0$, this aligns with our expectation that abandoning technology neutrality means more subsidies are awarded to FTTP projects.

5.2 Divergence in AltNet Coverage

Before we move onto the continuous DiD model, we run the S&A model one last time using AltNet footprint (% area of the region covered by AltNets) as the dependent variable. While the previous models document the divergence in network speeds, analysing AltNet footprint isolates a competitive mechanism by testing whether the intervention specifically deterred the entry of third-party providers.

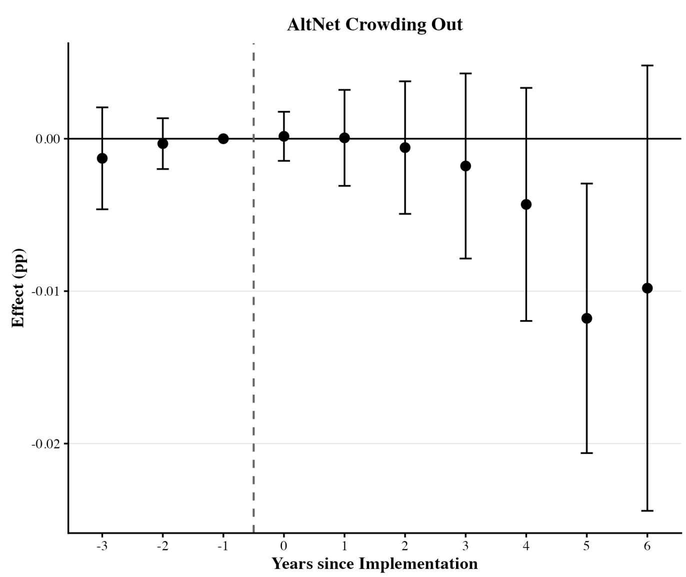


Figure 8: AltNet Mechanism Divergence

As shown in Figure 8, there is a decline in AltNet investment in treated areas that becomes significant by year 5. This timing aligns with the Full Fibre gap observed in Figure 7 Model (2A), suggesting that early state intervention for FTTC-based infrastructure reduced the commercial appeal of these regions to new entrants over time, this dynamic strongly motivates the presence of a copper trap mechanism.

5.3 Existence of Copper Trap

To test this hypothesis, Table 3 presents the continuous DiD estimates, the key parameter here is β_3. As discussed in the empirical framework, this measures the difference in the partial effect of a unit of $y_{SF,t-1}$ coverage on $y_{FF,t}$, between treated and untreated regions. Earlier we discussed the copper trap mechanism applying to regions in 'Zone 3'. If this mechanism holds, the penalty of the treatment to full fiber coverage should not be static; it should worsen the more that local market is saturated with FTTC.

Table 3: Copper Trap Interaction DiD Model

	FF: Strict (Year FE)	FF: Loose (No Year FE, Mat)	FF: Loose (No Year FE, Unm)
Post	0.053*	-0.022	-0.083^{***}
	(0.021)	(0.025)	(0.024)
$y_{SF,t-1}$	0.151***	0.023	-0.024
	(0.026)	(0.033)	(0.034)
GDHI per head (£)	0.008	0.059***	0.014**
	(0.009)	(0.008)	(0.004)
Residents Aged 65+ (%)	-6.405^{***}	-6.355^{***}	-5.214^{***}
	(1.110)	(1.199)	(1.092)
Exchange Density	-1.340	-2.285^{*}	-2.949^{***}
	(0.865)	(1.107)	(0.639)
Post × $y_{SF,t-1}$	-0.092^{**}	-0.091^{**}	0.001
	(0.029)	(0.034)	(0.033)
Linear Time Trend		0.015**	0.035***
		(0.005)	(0.004)
Num.Obs.	1378	1378	1872
R-squared	0.785	0.657	0.625
Local Authority Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	No	No
Cluster SE (Local Authority)	Yes	Yes	Yes

$^{+}p < 0.1, ^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001$

Dependent Variable is FF Coverage. GDHI, Density and %65+ are covariates in the vector X_t

That would imply the interaction coefficient (β_3) is negative, and indeed our results in Table 3 confirm this: β_3 is significant at -0.092 ($p < 0.01$). We plot the total treatment effect for a treated region, $\beta_1 + \beta_3 \times y_{SF,t-1}$, in Figure 9

The X-axis ($y_{SF,t-1}$) tracks the transition from market with zero existing FTTC infrastructure ($y_{SF,t-1} = 0$) to a fully state-subsidised FTTC market ($y_{SF,t-1} = 1$). The BDUK intervention

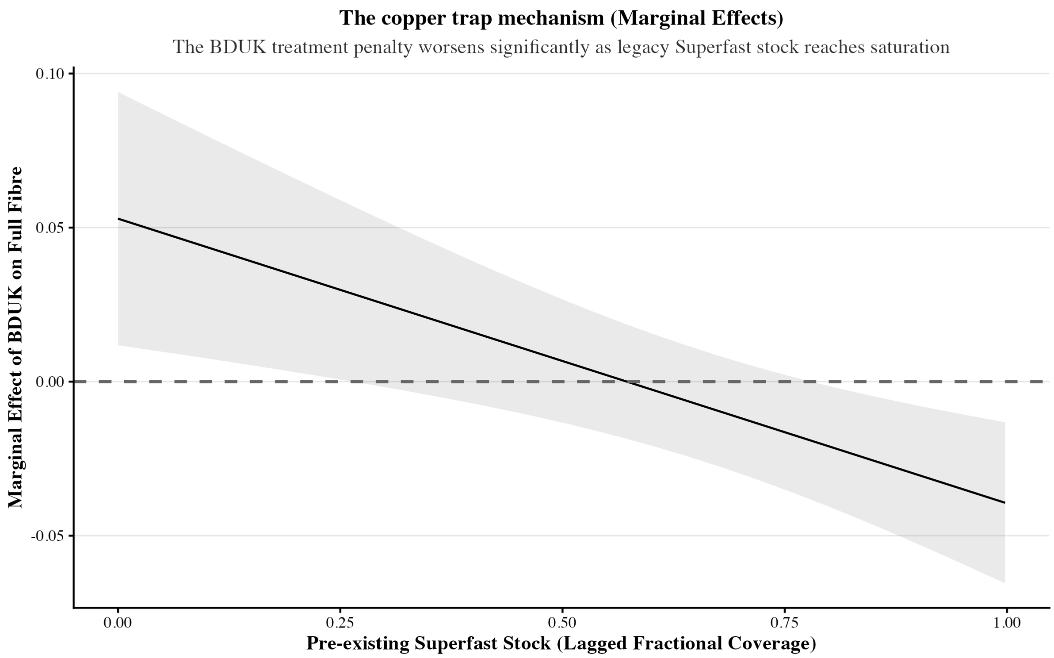


Figure 9: The copper trap mechanism (Marginal Effects)

provides an initial baseline boost of +0.053 to unserved areas—meaning the subsidy generates 5.3 percentage points *more* Full Fibre than an untreated market would deliver. However, this benefit is offset by an interaction penalty that scales with existing FTTC coverage. At 100% saturation, this penalty reaches its maximum of −0.092, reflecting the disincentives of building over FTTC lines. In a market with full FTTC saturation, this results in a negative 3.9 percentage-point treatment effect ($\beta_1 + \beta_3 \times 1.00 = 0.053 - 0.092 = 0.039$).

A concern with using TWFE for staggered rollouts is that if interventions are concentrated in specific years (such as the 2012–2014 cluster visible in Figure 4), Year Fixed Effects (TFE) might aggressively absorb the temporal variation, 'washing out' the true treatment effect. To address this, we run it again without Year FEs (Table 3, column 2). When Time Fixed Effects are relaxed and replaced with a linear time trend ('Loose' specification), the interaction penalty does not change, moving from -0.092 to -0.091 and still retains its significance.

Separately, note that the use of matching is validated in the final column of Table 3. When the Continuous Interaction model is run on the Unmatched sample, the interaction penalty vanishes (+0.001). The unmatched sample is polluted by hyper-dense urban outliers where private entrants can profitably overbuild legacy copper with Full Fibre, regardless of state intervention. By including these areas, the model averages away the above observed copper trap effect.

6 Discussion

The emergence of this divergence in broadband quality is the consequence of a flawed procurement design. A technology-neutral 30 Mbps threshold permitted BT to leverage legacy assets to undercut entrants, validating "asset sweating" concerns raised by Briglauer (2014). This market foreclosure culminated in 2013 when Fujitsu, the only major competing bidder with a nationwide full-fibre rollout plan, abruptly withdrew. The firm conceded that the government's commercial conditions made competing against BT's legacy copper monopoly unviable (National Audit Office 2013). This suppressed the inter-platform competition identiied by Distaso et al. (2006) as the primary driver of quality upgrades. Consultations with industry further warned that the gap-funding framework skewed the market toward the incumbent, whose lower weighted average cost of capital (WACC) starved AltNets of the ability to compete as bids were selected based on lowest NPV funding requirement (Sky UK 2014).

Dixit and Pindyck (1994) posit that firms treat deployment as a "call option" —rationally delaying investment until market uncertainty resolves. Analagously, our findings suggest that state subsidies devalued this option. The "prize" for a new provider is no longer the absolute demand for broadband, but merely the marginal willingness-to-pay for a speed upgrade. This compresses the expected value of the option and makes it difficult for a potential FTTP project to clear the old strike price, trapping the market in the FTTC status quo. This would not be a problem if consumers were willing to pay a sizeable premium to jump from 30 Mbps to 1,000 Mbps. But evidence from Ahlfeldt et al. (2014) on house prices demonstrate that while an initial upgrade to Superfast broadband commands a 2.8% price premium, the premium from an upgrade from "Superfast" to "Ultrafast" is less than 1%.

By delivering a "good enough" subsidised connection, the state rendered consumer demand for higher speeds highly inelastic so the marginal revenue available to a Full Fibre entrant collapsed. In the language of the earlier model it shifted the viability threshold from θ^* to a new higher θ^{**} , foreclosing any markets in that region to organic entry of FTTP providers.

These findings invite a broader consideration for the impact on social welfare. There is a tradeoff, and in that way the BDUK policy may be viewed as a second-best optimum (Laffont and Tirole 2000). Facing strict budgetary constraints and a high marginal cost of public funds, an FTTC rollout may have maximised overall welfare by connecting a vast number of rural households to basic fast internet while avoiding the larger subsidies required for FTTP — an investment whose external benefits may not have exceeded the high cost of public funds at the time.

In executing this strategy, policymakers prioritised immediate connectivity over the long-term opportunity cost of foregone full fibre capacity. While this preference represents a defensible choice, our reading of institutional appraisals suggests that the dynamic cost of lock-in was not actually considered during the programme's inception.

Over the long run, the consequence of this decision is that in many treated regions, state intervention delayed the evolution of the network further than if they had just abstained. Now as BDUK shifts toward universal gigabit coverage — through the £5 billion Project

Gigabit initiative — there is a risk of triggering another foreclosure effect. Locking in a single technological standard today may again crowd out unforeseen future technologies, such as advanced low-earth-orbit satellite networks or next-generation wireless architectures.

This is not unavoidable, one idea is to redirect subsidies toward passive, upgradable infrastructure — such as dark fibre backhaul and open-access ducting — rather than active electronics like cabinets (Krämer and Schnurr 2014). Subsidising this "deep" infrastructure lowers the initial capital barrier for all future providers which helps preserve the real option value of subsequent upgrades (Dixit and Pindyck 1994).

6.1 Back-of-the-Envelope Cost

During the window of Full Fibre divergence ($T = 4$ to $T = 6$), treated authorities exhibited an average Superfast stock (y_{SF}) of approximately 65%. Applying this to our continuous DiD model establishes an effective policy-induced reduction of full fibre coverage of roughly -0.06 (-0.092×0.65). This suggests the BDUK intervention locked out approximately 300,000 rural premises ($5,000,000 \times 0.06$) from receiving commercial Full Fibre that would have otherwise occurred in an untreated counterfactual world.

Based on standard deployment costs of £600 to £1,000 per premise passed (Ipsos MORI 2018), the fiscal burden required to rectify this policy-induced 'digital divide', ranges between £180 million and £300 million. This expenditure is relevant as it is being absorbed by the successor £5 billion "Project Gigabit" initiative which now aims for universal full fiber coverage. Effectively, the state is paying contractors a second time round to overbuild the very infrastructure they set up a decade prior. We conclude that while the early BDUK programme claimed a high Benefit-Cost Ratio based on immediate Superfast gains, this metric is overstated as it failed to account for this longer-term cost.

The UK is not unique in experiencing this unintended consequence. Recent evaluations of the French Broadband Plan by Bourreau et al. (2025) provide international corroboration. France also utilised state aid to rollout NGA broadband to white areas, by estimating a structural model of fiber entry, Bourreau et al. (2025) discovered that in 36% of cases, public subsidies were allocated to municipalities where private investment would have occurred anyway within three years, half of which would have been full fibre offerings - this result mirrors the penalty to full-fibre rollout in treated regions observed in our UK estimates.

7 Robustness

7.1 Staggered Bias Diagnostics

To validate the necessity of the Sun & Abraham estimator, we conducted a Goodman-Bacon decomposition on the standard TWFE model. The decomposition reveals that the standard TWFE model underestimates the true positive ATT. Specifically, TWFE implicitly uses already-treated Local Authorities as controls for newly treated ones. As Figure 10 demonstrates, these 'later vs. earlier treated' comparisons yield biased negative estimates.

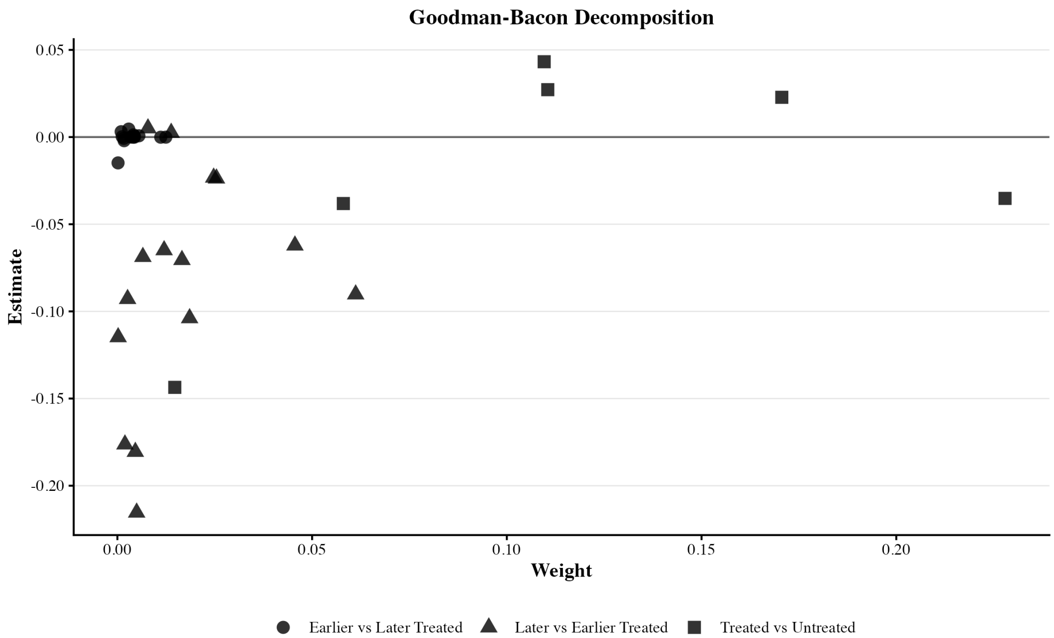


Figure 10: Goodman-Bacon Decomposition

This drags the overall estimate down, and the flaw necessitates our use of the interaction-weighted Sun and Abraham (2021) estimator. Though note these 'unclean comparisons' do have a small weight, so the TWFE model used for the continuous interaction estimates should only have a small bias, as most weight is applied on clean 'Treated vs. Never-Treated' comparisons.

7.2 Alternative Matching Algorithm

To ensure the copper trap interaction is a structural feature of the policy rather than an artifact of our Matching protocol, we re-estimate the Continuous Interaction DiD model across alternative matching algorithms by replacing our original sample with control groups derived via 1:1 Nearest Neighbor (without replacement) and Mahalanobis Distance matching.

Table 4: Robustness to Matching Algorithm Selection

	Genetic (Baseline)	Nearest Neighbor	Mahalanobis
Post	0.05*	0.07**	0.06*
	(0.02)	(0.02)	(0.02)
Lagged Superfast Stock	0.15***	0.18***	0.18***
	(0.02)	(0.02)	(0.02)
GDHI per head (£)	0.01+	0.01***	0.01***
	(0.00)	(0.00)	(0.00)
Residents Aged 65+ (%)	-6.40***	-6.71***	-6.78***
	(0.54)	(0.45)	(0.45)
Exchange Density	-1.34+	-2.20***	-2.19***
	(0.70)	(0.37)	(0.37)
Post × Lagged SF	-0.09***	-0.10***	-0.09***
	(0.03)	(0.02)	(0.03)
Num.Obs.	1378	1859	1859
R2	0.785	0.798	0.798
Local Authority Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Cluster SE (Local Authority)	Yes	Yes	Yes

- $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Coefficients represent the interaction term. Standard errors clustered by Local Authority.

As demonstrated in Table 4, β_3 remains stable across all specifications.

7.3 Alternative Estimators

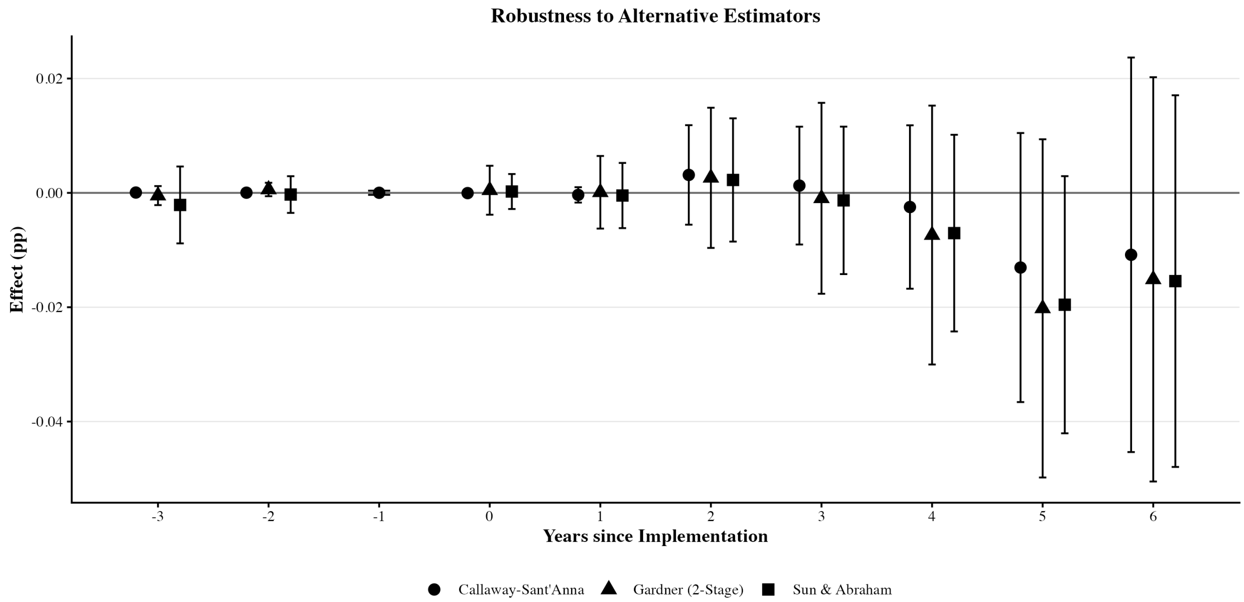


Figure 11: Robustness to Alternative Estimators

Using Figure 11, we show that our findings are invariant to the choice of modern heterogeneity-robust estimators. Re-estimating the baseline models using Borusyak, Callaway-Sant'Anna, and Gardner (2-Stage) estimators all confirm the same statistically significant downward structural trajectory for Full Fibre; estimates remain tightly bound.

7.4 Placebo Tests

To ensure our matching covariates are not themselves being affected by the treatment we run placebo tests on wealth (GDHI) and coaxial cable (Virgin Media) in Figure 12. Both our placebo tests yield a flat zero line, which corroborates our prior statement that these are invariant.

Furthermore, the fact the coefficients in all the event studies are close to zero lends support to our parallel trends assumption as it shows that treatment and control local authorities were both following similar trends in wealth and cable footprint.

Finally in panel (c), a Monte Carlo placebo test (shuffling treatment years 500 times) confirms the observed coefficient in our copper trap model has less than a 5% probability of being generated by some random permutation of the data.

7.5 Residual Plots

Below we display the residuals of the primary Full Fibre specification plotted against fitted values. The distribution has a cloud-like distribution around zero except for a cluster of points that form a line with a slope of -1, passing through the origin.

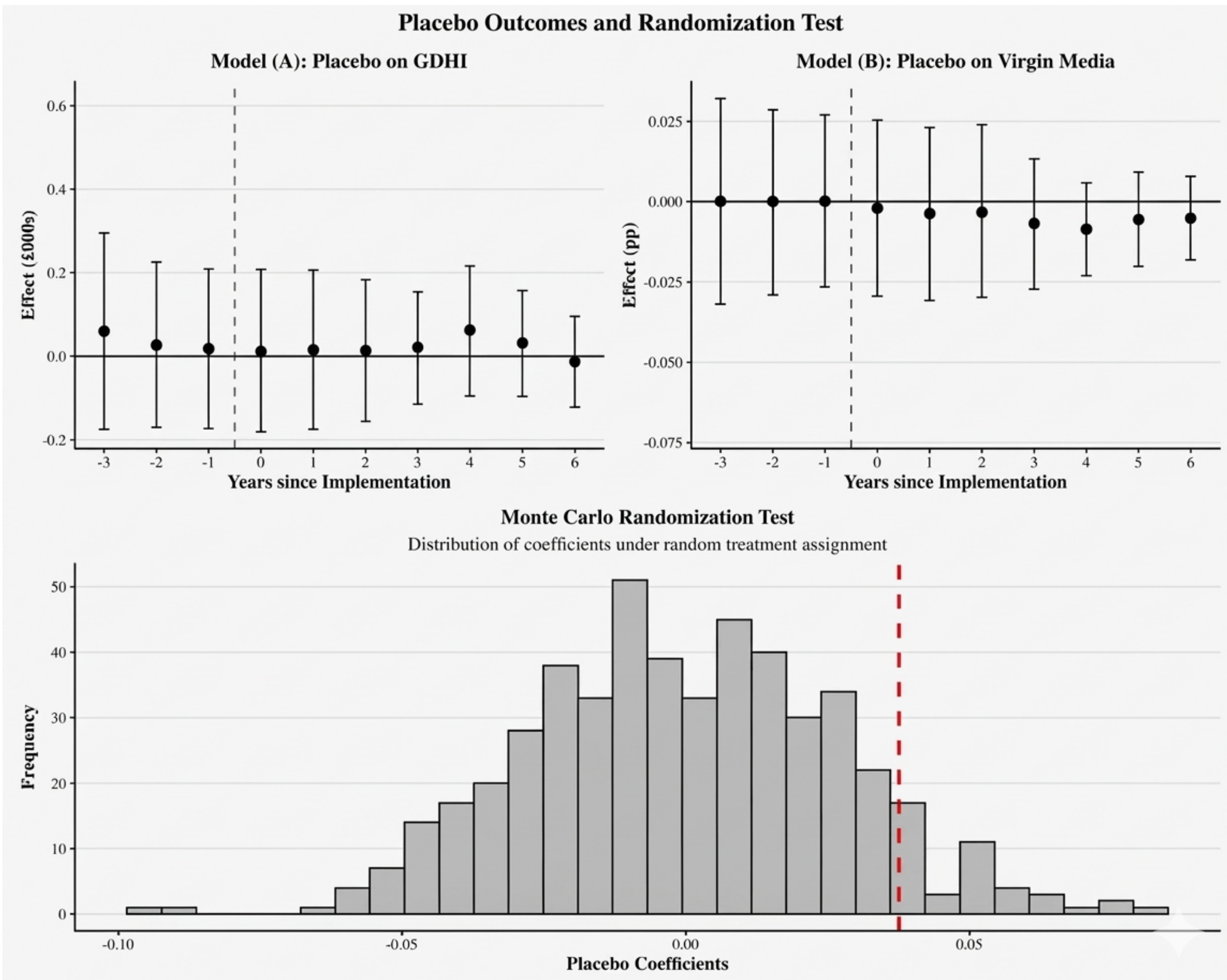
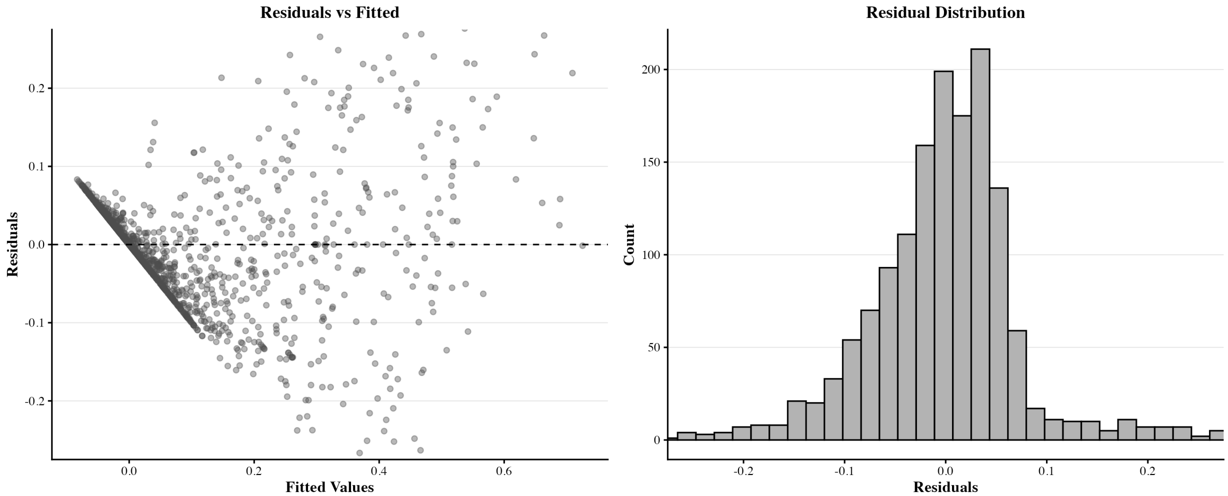


Figure 12: Placebo Outcomes and Randomization Test



This is an artifact of the data's zero-bound. Full Fibre coverage in many authorities was exactly 0 for about the first half of the panel, so our linear model mechanically generates residuals of $-\widehat{y}_{it}$ for these observations, inducing conditional heteroskedasticity.

While bounded data typically motivates non-linear specifications (such as Tobit or Fractional Response), applying maximum likelihood estimation to a short panel ($T = 16$) with high-dimensional unit fixed effects introduces the Incidental Parameter Problem (Neyman and Scott 1948). In non-linear models, the inability to partial-out fixed effects causes the estimation error of the incidental parameters to severely bias the structural treatment coefficients.

Consequently, we maintain a Linear Probability Model to ensure consistent parameter estimation, but to address this induced heteroskedacity we relied on heteroskedacity robust standard errors, clustered at the Local Authority level for all regressions.

8 Conclusion

This dissertation provides an illustration of the tension between political deliverables and long-term goals. By constructing a sequential investment game, we demonstrated how subsidising an intermediate technology compresses the marginal revenue available for subsequent upgrades. We then supported this idea that subsidised FTTC compresses the return on rolling-out FTTP by showing that (i) subsidies did cause a significant improvement in FTTC coverage using Sun and Abraham (2021) event study plots and, (ii) Local Authorities where FTTC penetrated deepest suffered the most severe reduction in subsequent FTTP upgrades. Finally, an ancillary result (iii) revealed that abandoning the "technology neutral" method in Phase 3 of actively began reversing these effects.

Consequently, we urge policymakers to abandon technology-neutrality in broadband. To safeguard further network upgrades, this will mean being more accepting of higher upfront fiscal burdens to procure a futureproofed technology. Although Phase 3 of the BDUK program eventually shifted toward an FTTP mandate, the legacy of Phases 1 and 2 is entrenched, and has left early-intervention areas stranded on the wrong side of a new digital divide.

We point out a couple avenues for further research. First, comparative analysis offers a way to evaluate the efficacy of the gap-funded public-private partnership. Cross-sectional studies of how Japan and South Korea avoided the copper trap to achieve universal FTTP penetration—provide a clear alternative (Kushida and Oh 2007). Contrasting the UK's reliance on incumbent-led PPPs with a more direct, technology-specific approach may help identify which policy regime best facilitates the transition to next-generation infrastructure.

Second, there is scope to investigate the broader consequences of 'technology neutrality' in mechanism design. Examining its role in 5G spectrum allocation (Cave 2010), renewable energy subsidies (Azar and Sandén 2011), and carbon policies (Kalkuhl et al. 2012) might clarify under what conditions this approach genuinely fosters innovation. Our tentative position is that neutrality may be optimal to foster innovation among nascent, competing technologies (such as different carbon capture methods). Applying it to a mature hierarchy where the differentiation is clear, such as FTTP, instead slows down adoption of that frontier.

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Appendix

Appendix A: Ofcom Data vs ThinkBroadband

Table 5: Ofcom Regulatory Data vs. ThinkBroadband Commercial Data

Feature	**Ofcom**	**Thinkbroadband**
Data Source	Statutory operator submissions	Independent discovery via speed-tests, and direct tracking of 100+ network footprints.
Unit	Excludes ”child” premises (e.g., individual apartments).	Includes ”child” premises, higher granularity.
Frequency	Quarterly	Daily
Method	Relies on internally modeled data from providers i.e. their maximal advertised speeds.	Primarily from site visitor's speed tests and independent audits. Secondarily augmented by modelling physical networks to account for attenuation.

Appendix B: Derivation of the SPNE

This Appendix provides the formal algebraic derivation of the thresholds established in Section 4.3. We consider the investment subgame at $t = 2$ following the demand shock δ.

Stage 2: The Post-Shock Investment Subgames ($t = 2$)

Firms observe θ_2 and simultaneously decide whether to invest. We evaluate the two potential subgames inherited from $t = 1$.

Subgame A: Untreated Counterfactual

Assume the State did not intervene at $t = 1$. The Entrant (E) evaluates building FTTP.

Because there is no existing high-speed network, E captures the absolute monopoly revenue. The Entrant invests if profits are non-negative ($\Pi_{E,F} \geq 0$):

$$\theta_2 \cdot r_F - K_F \geq 0$$

Solving for θ_2 establishes the Natural Monopoly Threshold for FTTP:

$$\theta^* = \frac{K_F}{r_F}$$

(Result: In a free market, any area reaching $\theta_2 \geq \theta^*$ at $t = 2$ organically receives FTTP).

Subgame B: Treated Reality

Assume the State intervened at $t = 1$, subsidising the Incumbent to build FTTC.

At $t = 2$, consumers already possess baseline quality, q_C.

If the Entrant (E) wishes to build FTTP, they face the Business Migration Effect (Bourreau et al. (2012)). They cannot extract absolute revenue r_F; they must convince consumers to switch, meaning they only extract the marginal willingness-to-pay for the speed upgrade: $(r_F - r_C)$. The Entrant's profit function compresses to:

$$\Pi'_{E,F} = \theta_2(r_F - r_C) - K_F \geq 0$$

Solving for θ_2 establishes the Overbuild Threshold for FTTP:

$$\theta^{**} = \frac{K_F}{r_F - r_C}$$

> **Proposition 1 (The Copper Trap Mechanism):** Because the incremental marginal revenue is strictly smaller than absolute monopoly revenue ($(r_F - r_C) < r_F$), the denominator shrinks. This mathematically dictates that $\theta^{**} > \theta^*$. The presence of state-sponsored FTTC elevates the commercial barrier to entry.

Stage 1: The Procurement Auction ($t = 1$)

At $t = 1$, viability θ_1 is low. The State identifies "White Areas" where investment is entirely absent ($\theta_1 r_C < K_{I,C}$).

Operating under a fixed budget, the State acts as a short-term cost-minimiser. It initiates a technology-neutral gap-funding auction for a minimum speed of q_C. Firms bid their required subsidy (S) to break even: $S = K - \theta_1 r$.

- The Incumbent bids FTTC: $S_{I,C} = K_{I,C} - \theta_1 r_C$
- The Entrant bids FTTP: $S_{E,F} = K_F - \theta_1 r_F$

Because the Incumbent reuses legacy copper, $K_{I,C} \ll K_F$. Therefore, $S_{I,C} < S_{E,F}$.

> **Proposition 2 (The Procurement SPNE):** The State strictly selects the Incumbent's FTTC bid. The SPNE of the entire game dictates that the State always subsidises FTTC *at* $t = 1$, mechanically forcing all White Areas into Subgame 2B at $t = 2$.